Subnetting Design & IP Addressing Plan Assignment

Comprehensive Long-Form Technical Report & Practical Solutions (Tasks 1 to 4)

Task 1: Subnetting 192.168.10.0/24 into 4 Equal Subnets

Given the base Class C network address **192.168.10.0/24**, we need to divide it into **4 equal-sized subnets**.

Mathematical Derivation:

- **1. Calculate Borrowed Subnet Bits (n):**
  To create 4 subnets: $2^n = 4 \implies n = 2 ext{ bits}$.
- **2. Calculate New CIDR Subnet Prefix:**
  Original prefix = 24 bits. New prefix = $24 + 2 = \mathbf{/26}$.
- **3. Calculate Subnet Mask:**
  In binary: `11111111.11111111.11111111.11000000` = `255.255.255.192`.
- **4. Calculate Block Size (Increment Interval):**
  Block Size = $256 - 192 = \mathbf{64}$ addresses per subnet.
- **5. Hosts Per Subnet:**
  Total IP addresses = 64. Usable host IPs = $64 - 2 = \mathbf{62 ext{ usable hosts}}$.

Subnetting Table for the 4 Equal Subnets:

Subnet #	Subnet Name / Role	Network Address	Subnet Mask	Usable Host IP Range	Broadcast Address
Subnet 1	Admin Network	`192.168.10.0`	`255.255.255.192`	`192.168.10.1` – `192.168.10.62`	`192.168.10.63`
Subnet 2	Staff / Workstation	`192.168.10.64`	`255.255.255.192`	`192.168.10.65` – `192.168.10.126`	`192.168.10.127`
Subnet 3	Guest Wi-Fi	`192.168.10.128`	`255.255.255.192`	`192.168.10.129` – `192.168.10.190`	`192.168.10.191`
Subnet 4	Server / Lab Network	`192.168.10.192`	`255.255.255.192`	`192.168.10.193` – `192.168.10.254`	`192.168.10.255`

Task 2: Device IP Address Allocation for Subnet 1 (192.168.10.0/26)

We select **Subnet 1** (Network Address: `192.168.10.0/26`, Usable Range: `192.168.10.1` to `192.168.10.62`) to allocate valid static IP addresses to 5 network devices. Note that `192.168.10.0` (Network ID) and `192.168.10.63` (Broadcast ID) are strictly excluded.

Device Allocation Breakdown:

Device Category	Device Description	Assigned IPv4 Address	Subnet Mask	Default Gateway
Router Interface	Main Gateway Interface (GigabitEthernet0/0)	`192.168.10.1`	`255.255.255.192`	`Self (192.168.10.1)`
Laptop 1	Admin Development Laptop	`192.168.10.2`	`255.255.255.192`	`192.168.10.1`
Laptop 2	Management Laptop	`192.168.10.3`	`255.255.255.192`	`192.168.10.1`
Network Printer	Office LaserJet Printer	`192.168.10.10`	`255.255.255.192`	`192.168.10.1`
Mobile Device	Admin Smartphone (Wi-Fi)	`192.168.10.20`	`255.255.255.192`	`192.168.10.1`

Task 3: Visual Subnet Plan & Device Connectivity Diagram

Below is a structural layout diagram showing the division of `192.168.10.0/24` into 4 equal subnets and the exact connection layout for devices on **Subnet 1**:

🌐 NETWORK SUBNET PLAN DIAGRAM (192.168.10.0/24 DIVIDED INTO 4 SUBNETS)

🔹 SUBNET 1: Admin Network (Active Topology)

- **Network ID:** 192.168.10.0/26
- **IP Range:** 192.168.10.1 – 192.168.10.62
- **Broadcast ID:** 192.168.10.63

Connected Devices (via Central Switch):
├─ 🌐 **Router (Gateway):** 192.168.10.1
├─ 💻 **Laptop 1:** 192.168.10.2
├─ 💻 **Laptop 2:** 192.168.10.3
├─ 🖨️ **Printer:** 192.168.10.10
└─ 📱 **Mobile:** 192.168.10.20

🔹 SUBNET 2: Staff Network

- **Network ID:** 192.168.10.64/26
- **IP Range:** 192.168.10.65 – 192.168.10.126
- **Broadcast ID:** 192.168.10.127

🔹 SUBNET 3: Guest Wi-Fi

- **Network ID:** 192.168.10.128/26
- **IP Range:** 192.168.10.129 – 192.168.10.190
- **Broadcast ID:** 192.168.10.191

🔹 SUBNET 4: Server / Lab Network

- **Network ID:** 192.168.10.192/26
- **IP Range:** 192.168.10.193 – 192.168.10.254
- **Broadcast ID:** 192.168.10.255

Task 4: Beginner-Friendly Step-by-Step Explanation ("Subnetting Explained to a Friend")

Imagine you bought a large plot of land represented by `192.168.10.0` that can hold 256 room numbers (from 0 to 255). If you want to build **4 equal housing societies** on this land, here is how you do the math step-by-step:

Step 1: How many bits do we need to borrow?

In networking, everything works in powers of 2 ($2^1=2, 2^2=4, 2^3=8$). To get **4 subnets**, we need $2^2 = 4$, which means we must borrow **2 bits** from the host part of the IP address.

Step 2: Finding the New Subnet Mask

Our original address `192.168.10.0/24` had 24 ones in its subnet mask (`255.255.255.0`). Adding the 2 borrowed bits gives us $24 + 2 = \mathbf{26 ext{ bits}}$ (written as **/26**).
In binary, the last octet becomes `11000000`, which equals $128 + 64 = \mathbf{192}$.
So, our new Subnet Mask for all 4 subnets is **`255.255.255.192`**!

Step 3: Calculating Hosts Per Subnet & Block Size

Subtract the last number of the subnet mask from 256: $256 - 192 = \mathbf{64}$.
This magic number **64** is our block size. Each subnet gets a total block of 64 IP addresses!

However, in every subnet, two IP addresses are reserved and cannot be assigned to devices:
1. **Network Address (First IP):** Identifies the subnet itself.
2. **Broadcast Address (Last IP):** Used to send messages to all devices in that subnet simultaneously.

Therefore, Usable Hosts = $64 - 2 = \mathbf{62 ext{ usable IPs}}$ per subnet!

Step 4: Writing down the Subnet Ranges & Assigning Devices

We start counting from 0 and increment by 64:
• **Subnet 1:** Starts at `.0`. Broadcast is `.63`. Usable IPs: `.1` to `.62`.
• **Subnet 2:** Starts at `.64`. Broadcast is `.127`. Usable IPs: `.65` to `.126`.
• **Subnet 3:** Starts at `.128`. Broadcast is `.191`. Usable IPs: `.129` to `.190`.
• **Subnet 4:** Starts at `.192`. Broadcast is `.255`. Usable IPs: `.193` to `.254`.

For Subnet 1, we pick IP `.1` for our Router (Gateway), `.2` and `.3` for Laptops, `.10` for the Printer, and `.20` for the Mobile device. This keeps our network organized, isolated, and secure!